

Report on Rank Classification and Canonical Forms over Prime Fields

Math 484 - Introduction to Quantum Algorithms &
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1 Introduction

The research of tensor ranks and canonical forms are considered important because the analogous results have direct connections and implications in many areas of research including but not limited to quantum information theory, quantum mechanics, signal processing, electrical engineering.

While many of the classical research focuses on infinite fields such as real or complex numbers, papers that we are considering are focused on finite fields, specifically on prime fields $\mathbb{F}_p$. Working over finite fields helps eliminate computational barriers that are encountered while using infinite fields.

This report summarizes the problems, methods, and the direct results of three papers “Canonical forms of $2 \times 2 \times 2$ and $2 \times 2 \times 2 \times 2$ arrays over $\mathbb{F}_2$ and $\mathbb{F}_3$ ” by Murray R. Bremner and Stavroz G. Stavrou, “Canonical forms of $2 \times 2 \times 2$ and $2 \times 2 \times 2 \times 2$ symmetric tensors over prime fields” by Stavroz G. Stavrou, and “Rank classification of $2 \times 2 \times 2 \times 2 \times 2$ tensors over $\mathbb{F}_2$ ” by Stavroz G. Stavrou, Richard M. Low, Nicholas J. Hernandez.

2 Preliminaries

All three papers give and use either the same almost identical definitions and notations. We will not repeat all, mentioned some briefly to make it easier to follow this report.

All papers focus on **n -dimensional arrays** $X = (x_{i_1}, \dots, x_{i_n})$ of size $2 \times 2 \times \dots \times 2$ (n factors) with entries from finite field $\mathbb{F}_p$. These are **tensor products** (or **outer products**) of n non-zero vectors of size 2. Here, n is the order of the array. Papers use the terms **multidimensional array**, **array**, or **tensor** interchangeably and so will we.

We will not give the exact definition of calculating the tensors from the tensor product or how they are represented, but we will mention the fact that all three papers define the same lex order based on the order of indices of the elements of the array. This total ordering allows defining a minimal element among a set of arrays which is defined to be the **canonical form** of that set.

Next we will give the notion of **rank** or **outer product rank**. For that we first define a **simple** tensor.

Definition: An order- n tensor is simple if and only if it can be written as the tensor product of n non-zero vectors.

$$X = u^{(1)} \otimes \dots \otimes u^{(n)}$$

Note that if one takes all the elements of the tensor product the same, then that tensor is said to be a **symmetric simple** tensor.

Now we can define the rank of a tensor.

Definition: Rank of an order- n tensor X is the minimum number of terms used while writing X as a sum of non-zero simple order- n tensors.

If one limits the simple tensors in the definition to symmetric simple tensors only, then that gives the **symmetric rank** of a tensor.

Lastly, we will mention the group actions and orbits without going into detail. Each paper defines a similar action group, the semi-direct product $GL_2(\mathbb{F}_p)^n \rtimes S_n$ where $GL_2(\mathbb{F}_p)^n$ is the direct product of the general linear group and S_n is the symmetry group. In the case of symmetric tensors, S_n has no effect. The orbits are defined to be the set of tensors that is closed under the action of the defined group. The important point here is that the action does not change the tensor rank, so all tensors in an orbit have the same rank.

In the following section we will go over each paper give details over the problem, methods, and the results.

3 “Canonical forms of $2 \times 2 \times 2$ and $2 \times 2 \times 2 \times 2$ arrays over $\mathbb{F}_2$ and $\mathbb{F}_3$ ” by Bremner and Stavrou

3.1 Problem Definition

The main goal of this research is to determine a complete classification of multi-dimensional arrays of two sizes: $2 \times 2 \times 2$ and $2 \times 2 \times 2 \times 2$ over the prime fields $\mathbb{F}_2$ and $\mathbb{F}_3$. The classification is conducted with respect to the action of the large symmetry group. The classification consists of determining the size, rank and the canonical form of each generated orbit.

3.2 Motivation

In this paper Bremner and Stavrou tackles this gap in the existing research which is relatively simpler compared to the case over infinite fields and can be reduced to an elementary problem of combinatorial enumeration[1] and can be solved via computer algebra.

Apart from being a problem of mathematical interest, the classification of multi-dimensional arrays –especially the analog case over infinite fields– along with the closely related topic of hyperdeterminants have direct connections to algebraic geometry and representation theory. Also, the results from these research areas have applications in other areas such as quantum information theory, numerical

analysis, and signal processing. The authors list some other areas and refer to other research on these connections see[1]

3.3 Methodology

As mentioned, the classification is done by partitioning the set of all arrays into disjoint orbits under the group action. The group action considered in this paper is the large symmetry group. The semi-direct product can be applied in two steps, the action of the general linear group and the action of the symmetry group.

The authors, being limited by the computational power and memory limitations of 2011, employ sophisticated programming techniques to tackle this problem. These computational constraints are the main reasons for this paper being limited order-3 and order-4 over $\mathbb{F}_2$ and $\mathbb{F}_3$. The application of the described algorithms yielded a complete and definitive classification of $2 \times 2 \times 2$ and $2 \times 2 \times 2 \times 2$ arrays over the prime fields $\mathbb{F}_2$ and $\mathbb{F}_3$. The authors present the maximum rank, number of arrays in each rank, and the size and rank of each orbit along with its canonical form.

The authors also provide pseudo-codes for each algorithm, for more detail see[1].

3.3.1 Order-3 Arrays

For the smaller $2 \times 2 \times 2$ case the authors use a “straightforward approach,” first classifying arrays by rank and then identifying the orbits within each rank.

- **Algorithm 1 (Rank Classification):** This algorithm uses an inductive method. It begins with the set of rank 0 (the zero array) and rank 1 arrays (the pure tensors). Assuming the arrays of rank r have already been computed, to find the arrays of rank $r + 1$ it computes all possible sums of rank r arrays and rank 1 arrays. Any array that has not been previously classified into a smaller rank.
- **Algorithm 2 (Orbit Classification):** This algorithm iterates through the set of each rank and selects the minimal array. Computes its entire orbit under the group action and removes all elements of that orbit from the set. This process is repeated until no arrays remain, yielding a complete list of canonical forms and their orbits.

3.3.2 Order-4 Arrays

Even though the *algorithms 1* and *2* can be used for the larger case of order-4, the authors state that they are not practical in terms of time constraints. They use more sophisticated algorithms and Maple programming techniques. They combine these techniques with the reverse approach of the first method by determining the orbits and then identifying the rank of each orbit.

- **Algorithm 3 (Spinning Algorithm):** This algorithm utilizes the fact that the general linear group $GL_2(\mathbb{F}_p)$ and the symmetry group S_n can be generated by two matrices and two permutations, respectively. It iteratively applies the generators of the symmetric group to the set of known orbit elements until no new elements are generated.
- **Algorithms 4 (Signal Arrays):** To overcome memory limitations this technique utilizes a large “signal array” initialized to zero and where each index corresponds to a flattened array. The algorithm finds the array corresponding to the first index with a zero, computes its orbit using the spinning algorithm and marks all arrays in that orbit with the orbit’s identifier. After all orbits are determined, the rank of each is determined using once again the signal arrays and the same idea of checking the sum of rank r array with rank 1 array. This time however, since the orbits are known, finding the rank of a single array in an orbit is enough.

3.4 Results

The application of the described algorithms yielded a complete and definitive classification of $2 \times 2 \times 2$ and $2 \times 2 \times 2 \times 2$ arrays over the prime fields $\mathbb{F}_2$ and $\mathbb{F}_3$. The authors present the maximum rank, number of arrays in each rank, and the size and rank of each orbit along with its canonical form.

3.4.1 Order-3 Arrays

Over $\mathbb{F}_2$ $2 \times 2 \times 2$ arrays have the maximum rank is 3. The distribution of arrays across ranks and their approximate percentages is as follows:

Rank	0	1	2	3
Count	1 array	27 arrays	162 arrays	66 arrays
%	0%	11%	63%	26%

Table 1: $2 \times 2 \times 2$ over $\mathbb{F}_2$

The authors noted that this distribution differs notably from that over the real numbers, where the percentages for ranks 0, 1, 2, and 3 are approximately 0%, 0%, 79%, and 21%, respectively. The classification under the large symmetry group yielded a total of 6 distinct orbits. For more details on the properties of each orbit, see[1].

Over $\mathbb{F}_3$ $2 \times 2 \times 2$ arrays have the maximum rank is 3. The distribution of arrays across ranks and their approximate percentages is as follows:

Rank	0	1	2	3
Count	1 array	128 arrays	4032 arrays	2400 arrays
%	0%	2%	61%	27%

Table 2: $2 \times 2 \times 2$ over $\mathbb{F}_3$

Under the large symmetry group, the classification yielded a total of 6 distinct orbits. For more details on the properties of each orbit, see[1].

3.4.2 Order-4 Arrays

Over $\mathbb{F}_2$ there are 65,536 arrays. The computations reveal a maximum rank of 6.

Rank	0	1	2	3	4	5	6
Count	1	81	2268	21744	37530	3888	24
%	0.0015%	0.1236%	3.4607%	33.1787%	57.2662%	5.9326%	0.0366%

Table 3: $2 \times 2 \times 2 \times 2$ over $\mathbb{F}_2$

The authors identified 30 distinct orbits under the action of the large symmetry group. For more detailed information on the orbits refer to [1].

Over $\mathbb{F}_3$ there are 43,046,721 arrays. The computations reveal a maximum rank of 5.

Rank	0	1	2	3	4	5
Count	1	512	101376	6397441	36018623	528768
%	0.000002%	0.001189%	0.235502%	14.861620%	83.673328%	1.228358%

Table 4: $2 \times 2 \times 2 \times 2$ over $\mathbb{F}_3$

The authors identified 49 distinct orbits under the action of the large symmetry group. For more detailed information on the orbits refer to [1].

4 “Canonical forms of $2 \times 2 \times 2$ and $2 \times 2 \times 2 \times 2$ symmetric tensors over prime fields” by Stavros G. Stavrou

4.1 Problem Definition

The main goal of this research is to investigate and do computations to determine canonical forms of symmetric tensors. The definitions of symmetric tensors and symmetric rank are given. Canonical forms are computed with respect to the action of general linear group $GL_2(\mathbb{F}_p)$ where p is prime. Then, symmetric rank of canonical forms, as well as orbits of $2 \times 2 \times 2$ and $2 \times 2 \times 2 \times 2$ symmetric tensors over prime fields $\mathbb{F}_p$ for $p = 2, 3, 5, 7, 11, 13, 17$ are computed using computer algebra.

4.2 Motivation

The analysis of tensors is an open area of research dating back to late 19th century. The notions of tensor rank and tensor decomposition followed after research of tensors started. Then, models computing these decompositions were formed and collectively referred as CPD (canonical polyadic decomposition).

Tensors are useful in many fields of technology. These include data mining, gene expression classification. Similarly, symmetric tensors are used in areas such as electrical engineering, signal processing.

For $2 \times 2 \times 2$ and $2 \times 2 \times 2 \times 2$ determining canonical forms of tensors over $\mathbb{R}$ and $\mathbb{C}$ has not been determined and this problem is related to quantum entanglement of four qubits. However, for $2 \times 2 \times 2$ and $2 \times 2 \times 2 \times 2$, symmetric tensors over $\mathbb{R}$ and $\mathbb{C}$ was solved by Gurevich[5]. The researchers has already published the results for determining canonical forms problem over $\mathbb{F}_2$ and $\mathbb{F}_3$. The aim of this paper is to expand this problem to prime fields $\mathbb{F}_p$.

4.3 Methodology

4.3.1 Order-3 Arrays

The set of $2 \times 2 \times 2$ tensors over the prime field $\mathbb{F}_p$ contains p^8 tensors because there would be p options for each eight entries in the tensor. For the symmetric tensor case, this number reduces to p^4 . The algorithm has two main elements:

Generate all the tensors in each rank and apply the group action so that the set of tensors in each rank is decomposed into a disjoint union of orbits.

The algorithm was used before by the authors for $\mathbb{F}_2$ and $\mathbb{F}_3$ but for classification of symmetric tensors they have made two modifications. Firstly, they set a rule that the set of rank 1 tensors must only contain symmetric simple tensors. For second, the same group element must act on every mode of the tensor of which its orbit is being to computed. The algorithm is as follows.

1. The only rank 0 tensor is the zero tensor.
2. The set of symmetric rank 1 tensors is $\{a \otimes a \otimes a \mid a \in F_{2p}\}$ where a is a non zero vector.
3. Assume that the tensors that have $rank_S(r)$ has been computed. Take all sums of $X + Y$ such that, $rank_S(X) = r$ and $rank_S(Y) = 1$. It can be concluded that, $rank_S(X + Y) \leq r + 1$.
4. Only retain those $X + Y$ that have: $rank_S(X + Y) = r + 1$.
5. Order these tensors within each symmetric rank in lexical order.
6. For each symmetric rank repeat the steps down below until no tensor is left in the current symmetric rank:
 - On the current symmetric rank, choose the minimal element of the set of tensors.
 - Compute the orbit of this element under the action of the symmetry group.
 - Remove the elements of this orbit from the set.

Order-4 Arrays

For $\mathbb{R}$ and $\mathbb{C}$: Weinberg [5] classifies canonical form of $2 \times 2 \times 2 \times 2$ symmetric tensors over $\mathbb{R}$ and $\mathbb{C}$, continuing from Gurevich. Over $\mathbb{R}$, there are nine canonical forms. Considering the group action $GL_{2,2,2,2}(\mathbb{C})$ there are six canonical forms.

For prime fields $\mathbb{F}_p$, the algorithm described above for $2 \times 2 \times 2$ generalizes nicely to $2 \times 2 \times 2 \times 2$, so it was used with small modifications. These modifications are:

- Instead of considering outer product of 3 non zero tensors, use outer product of four nonzero tensors.
- Add an extra index $I \in \{1, 2\}$. This allows group action on the fourth mode.

4.4 Results

4.4.1 Order-3 Arrays

$\mathbb{F}_2$: There are 256 tensors over $\mathbb{F}_2$, 16 of which are symmetric. The number of symmetric tensors in each symmetric rank and the approximate percentages are given in the table below. The percentages sum to 50%, the reason is only half of the tensors have a decomposition into a sum of symmetric simple tensors. The number of orbits is 4.

Rank	0	1	2	3
Count	1	3	3	1
%	6.25%	18.75%	18.75%	6.25%

Table 5: Symmetric Tensors over $\mathbb{F}_2$

$\mathbb{F}_3$: There are 6561 tensors over $\mathbb{F}_3$, 81 of which are symmetric. Unlike $\mathbb{F}_2$, all symmetric tensors have a decomposition into a sum of symmetric simple tensors. The number of orbits is 6.

Rank	0	1	2	3	4
Count	1	8	24	32	16
%	1.23%	9.88%	39.51%	29.63%	19.75%

Table 6: Symmetric Tensors over $\mathbb{F}_3$

$\mathbb{F}_5$: There are 390,625 tensors over $\mathbb{F}_5$, 625 of which are symmetric. Maximum symmetric rank is 3. The number of orbits is 6.

Rank	0	1	2	3
Count	1	24	240	360
%	0.16%	3.84%	38.40%	57.60%

Table 7: Symmetric Tensors over $\mathbb{F}_5$

$\mathbb{F}_7$: There are 5,764,801 tensors over $\mathbb{F}_7$, 2,401 of which are symmetric. The number of orbits is 14.

Rank	0	1	2	3	4	5
Count	1	16	128	688	1232	336
%	0.04%	0.67%	5.33%	28.65%	51.31%	13.99%

Table 8: Symmetric Tensors over $\mathbb{F}_7$

$\mathbb{F}_{11}$: There are 214,358,881 tensors over $\mathbb{F}_{11}$, 14,641 of which are symmetric. The number of orbits is 6.

Rank	0	1	2	3
Count	1	120	6600	7920
%	0.01%	0.82%	45.08%	54.09%

Table 9: Symmetric Tensors over $\mathbb{F}_{11}$

$\mathbb{F}_{13}$: There are 815,730,721 tensors over $\mathbb{F}_{13}$, 28,561 of which are symmetric. The orbit number is 14.

Rank	0	1	2	3	4
Count	1	56	1568	16016	10920
%	0.00%	0.02%	5.49%	56.08%	38.23%

Table 10: Symmetric Tensors over $\mathbb{F}_{13}$

$\mathbb{F}_{17}$: There are 6,975,757,441 tensors over $\mathbb{F}_{17}$, 83,521 of which are symmetric. The orbit number is 6.

Rank	0	1	2	3
Count	1	288	39168	44064
%	0.00%	0.34%	46.9%	52.76%

Table 11: Symmetric Tensors over $\mathbb{F}_{17}$

For $p = 19$, there are nearly 17 billion tensors and over 130,000 of these tensors are symmetric. The authors note that for this p and larger primes the memory of the program was insufficient which is why they stopped at $p = 17$.

4.4.2 Order-4 Arrays

$\mathbb{F}_2$: There are 65,536 tensors over $\mathbb{F}_2$, 32 of which are symmetric. Similarly to $2 \times 2 \times 2$ case, all symmetric tensors cannot be expressed as a sum of symmetric simple tensors. Only 8 out of 32 can be expressed as such.

Rank	0	1	2	3
Count	1	3	3	1
%	3.125%	9.375%	9.375%	3.125%

Table 12: Symmetric Tensors ($2 \times 2 \times 2 \times 2$) over $\mathbb{F}_2$

$\mathbb{F}_3$: There are 43,046,721 tensors over $\mathbb{F}_3$, 243 of which are symmetric. Only 81 out of the 243 symmetric tensors can be expressed as a sum of symmetric simple tensors.

Rank	0	1	2	3	4	5	6	7	8
Count	1	4	10	16	19	16	10	4	1
%	0.41%	1.65%	4.12%	6.58%	7.82%	6.58%	4.12%	1.65%	0.41%

Table 13: Symmetric Tensors ($2 \times 2 \times 2 \times 2$) over $\mathbb{F}_3$

$\mathbb{F}_5$: There are 152,587,890,625 tensors over $\mathbb{F}_5$. 3,125 of which are symmetric. In $\mathbb{F}_5$, all symmetric tensors have a symmetric decomposition.

Rank	0	1	2	3	4	5	6	7	8	9	10
Count	1	6	21	56	126	249	395	570	690	660	360
%	0.03%	0.19%	0.67%	1.79%	4.03%	7.68%	12.64%	18.24%	22.08%	21.12%	11.52%

Table 14: Symmetric Tensors ($2 \times 2 \times 2 \times 2$) over $\mathbb{F}_5$

$\mathbb{F}_7$: There are 33,232,930,569,601 tensors over $\mathbb{F}_7$, 16,807 of which are symmetric. All symmetric tensors have a symmetric decomposition.

Rank	0	1	2	3	4	5	6
Count	1	24	276	19132	7119	6615	840
%	0.01%	0.14%	1.64%	11.50%	42.38%	39.36%	5.00%

Table 15: Symmetric Tensors ($2 \times 2 \times 2 \times 2$) over $\mathbb{F}_7$

5 "Rank classification of 2 x 2 x 2 x 2 x 2 tensors over $\mathbb{F}_2$ " by Stavroz G. Stravrou, Richard M.Low, Nicholas J. Hernandez

5.1 Problem Definition

The primary focus of this research is to determine the rank classification of $2 \times 2 \times 2 \times 2 \times 2$ tensors over $\mathbb{F}_2$ utilizing computing techniques. The definitions of an order n -tensor, simple tensor, and tensor rank are given. Canonical forms of the maximum rank tensors are computed under the symmetry group $GL_2(\mathbb{F}_2)^5 \rtimes S_5$. Then, the maximum tensor rank, which is found to be 9, as well as the 261 canonical forms of the maximum rank tensors, are computed using computer algebra

5.2 Motivation

The programming techniques previously used for computing $2 \times 2 \times 2$ and $2 \times 2 \times 2 \times 2$ (order 3 and 4) by Bremner and Stravrou break down completely when using larger tensor formats, specifically memory usage and run-time, due to the vast number of operations being the biggest obstacles to overcome.

The core goal of the research was to determine the maximum tensor rank for the specific format of $2 \times 2 \times 2 \times 2 \times 2$ tensors over the finite field $\mathbb{F}_2$. Through the new programming techniques and computation, the maximum rank was found to be 9. By finding the maximum rank for the Order-5 case, the authors were enabled to prove that the maximum rank of an order- n tensor of format $2 \times 2 \times \dots \times 2$ ($n \geq 5$) over $\mathbb{F}_2$ is bounded by $9 \cdot 2^n - 5$.

The research also focused on classifying canonical forms for the maximum rank tensors. The goal was to compute the minimal element of each orbit under lexical ordering for the rank 9 under the symmetry group $GL_2(\mathbb{F}_2)^5 \rtimes S_5$, which resulted in the determination of 261 canonical forms.

5.3 Methodology

5.3.1 Computing Tensors in Each Rank

They approached the problem firstly by partitioning the set of tensors into disjoint sets, R_j , based on their tensor rank.

A Boolean hash table with 2^{32} distinct values is initialized to false. This table tracks which tensors have already been classified to ensure the sets R_j are disjoint.

The set of rank 1 tensors (R_1) are generated by considering the simple tensors. This yielded 243 tensors.

To generate higher rank tensors n (R_n), the rank 1 tensor ($x \in R_1$) and a rank $n - 1$ tensor ($y \in R_{n-1}$) are iteratively summed modulo 2:

$$Z = (x + y) \pmod{2}$$

If the resulting tensor Z had not yet occurred (hash table value is false), its value in the hash table was set to true, and Z was added to the list R_9 .

This iterative process continued until the maximum rank was reached, ensuring that every tensor was classified into one and only one rank set.

5.3.2 Computing Canonical Forms of Maximum Rank Tensors

After the maximum rank tensors (R_9) were identified, the canonical forms are computed using a two-step partitioning process based on the symmetry group $GL_2(\mathbb{F}_2)^5 \rtimes S_5$.

Algorithm 1 - Partition by $GL_2(\mathbb{F}_2)^5$

1. Choose the smallest unclassified element of R_9 .
2. Compute its orbit under the action of $GL_2(\mathbb{F}_2)^5$.
3. All elements of this orbit are removed from R_9 .
4. The smallest element in the orbit (under lexical ordering) is the orbit representative with respect to $GL_2(\mathbb{F}_2)^5$.

Algorithm 4 - Partition by S_5

1. The disjoint sets found in Step 1 are being further partitioned by the action of the permutation group.
2. The minimal element of these final orbits now represents the canonical form of the maximum rank tensors under the combined group action $GL_2(\mathbb{F}_2)^5 \rtimes S_5$.

5.4 Results

5.4.1 Maximum Rank and Classification

The complete rank classification of all 2^{32} tensors of format $2 \times 2 \times 2 \times 2$ over $\mathbb{F}_2$ was successfully achieved and for this format it was found to be 9. The distribution of all 2^{32} tensors across ranks and their approximate percentages is as follows:

Rank	Count	Approx. %
Rank 0	1	$2.32 \times 10^{-8}\%$
Rank 1	243	$5.66 \times 10^{-6}\%$
Rank 2	24,948	$5.81 \times 10^{-4}\%$
Rank 3	1,351,836	$3.15 \times 10^{-2}\%$
Rank 4	39,365,190	0.92%
Rank 5	545,193,342	12.96%
Rank 6	2,398,267,764	55.84%
Rank 7	1,299,295,404	30.25%
Rank 8	11,460,744	$4.22 \times 10^{-2}\%$
Rank 9	7,824	$1.82 \times 10^{-4}\%$

Table 16: Rank Classification of $2 \times 2 \times 2 \times 2 \times 2$ tensors over $\mathbb{F}_2$

5.4.2 Canonical Forms

There are 261 canonical forms in total for the maximum rank (rank 9) tensors. These forms are the minimal elements of the orbits under the final symmetry group action.

The set of rank 9 tensors was initially partitioned into two large disjoint orbits under the $GL_2(\mathbb{F}_2)^5 \rtimes S_5$ action, with sizes 48 and 7,776. The orbit of size 48 split into 9 disjoint orbits under the action of S_5 . The orbit of size 7,776 split into 252 disjoint orbits under the action of $GL_2(\mathbb{F}_2)^5 \rtimes S_5$. The table below shows The orbit of size 48 split into 9 disjoint orbits under the action of S_5 .

Orbit Size	Canonical Form
5	01101011101111011011110111010110
10	01101011101111011101011001101011
1	01111110111010011110100110010111
1	10010111011111100111111011101001
5	10010111011111101110100110010111

Table 17: Subset of Canonical Forms

5.4.3 New Upper Bound

As a direct corollary of finding the maximum rank to be 9 for the order-5 case, a new upper bound for higher-order tensors is established: For order- n tensors of format $2 \times \dots \times 2$ where $n \geq 5$, the maximum rank is bounded above by $9 \cdot 2^n - 5$.

6 Conclusion

In this report we considered three research on classification of tensors over prime fields and the sophisticated programming techniques used. The ranks in each order and prime field were calculated along with their canonical forms. All three of the research previously were greatly limited by the computational power and memory limitation constraints of their time and the programming techniques known at the time. The progress through years over each research shows that as the computational capacity increases more results can be obtained in future work.

7 References

References

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